Bayes Estimation

Outline

1) Bayes risk, Bayes estimator
2) Examples
3) Conjugate priors

Frequentist Motivation

Model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ for data X

Loss $L(\theta, d)$, Risk $R(\theta; \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))]$

The <u>Bayes risk</u> is the average-case risk, integrated wrt some measure Λ, called <u>prior</u>

For now, assume $\Lambda(\Theta) = 1$ (prob. meas.)

Later we will allow to be <u>improper</u> ($\Lambda(\Theta) = \infty$)

Note: • Λ and $c\Lambda$ for $c > 0$ functionally equiv.

• avg risk makes sense even if we don't "believe" $\theta \sim \Lambda$

$$r_\Lambda(\delta) = \int_\Theta R(\theta, \delta)\, d\Lambda(\theta)$$

$$= \mathbb{E}_{\theta\sim\Lambda}[\overbrace{R(\theta; \delta)}^{\mathbb{E}[L(\theta, \delta(X)) \mid \theta]}] \quad \text{where } \theta \sim \Lambda$$

$$= \mathbb{E}[L(\theta, \delta(X))] \quad \underbrace{\text{where } \theta \sim \Lambda,\ X \mid \theta \sim P_\theta}$$

$\mathbb{E}$ now means wrt <u>joint</u> distr. of (θ, X)

An estimator δ minimizing $r_\Lambda(\cdot)$ is called <u>Bayes</u> (a <u>Bayes estimator</u>). Dep. on $\mathcal{P}, \Lambda, L$

$$r_\Lambda(\delta) = \mathbb{E}[\overbrace{\mathbb{E}[L(\theta, \underbrace{\delta(X)}) \mid X]}^{\text{minimize one } X \text{ at a time}}]$$

Note: we choose this after seeing X

Prior, Posterior

Usual interp. of Λ is "prior belief about θ before seeing the data"

Conditional dist. $\Lambda(\theta \mid X)$ called posterior dist.
"belief after seeing the data"

Epistemic uncertainty:
"I think there is a 50% chance that..."
More on this next time

Densities: prior $\lambda(\theta)$, likelihood $p_\theta(x)$ or $p(x \mid \theta)$

Joint density $\lambda(\theta) p_\theta(x)$

Marginal density $q(x) = \int_\Theta \lambda(\theta) p_\theta(x) d\theta$

Posterior density $\lambda(\theta \mid x) = \dfrac{\lambda(\theta) p_\theta(x)}{q(x)}$

Bayes estimator depends on posterior:

$$\delta_\Lambda(x) = \arg\min_d \mathbb{E}[L(\theta, d) \mid X]$$

$$= \arg\min_d \int_\Theta L(\theta, d) \lambda(\theta \mid x) d\theta$$

Solve for Bayes estimator "one x at a time"

Bayes Estimator

Model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, (proper) prior Λ

Assume

- $L(\theta, d) \geq 0$
- $r_\Lambda(\delta_0) < \infty$ for some $\delta_0(x)$

Then $\delta_\Lambda(x)$ is Bayes, with $r_\Lambda(\delta_\Lambda) < \infty$

$$\text{iff } \delta_\Lambda(x) \in \underset{d}{\operatorname{argmin}}\ \mathbb{E}\left[L(\theta, d) \mid X = x\right] \quad \text{a.e. } x$$

$\mathbb{P}(\delta_\Lambda(x) \notin \operatorname{argmin}) = 0$
wrt marginal

Proof

($\Leftarrow$) δ any other estimator

$$\mathbb{E}\left[L(\theta, \delta(x)) \mid X\right] \overset{a.s.}{\geq} \mathbb{E}\left[L(\theta, \delta_\Lambda(x)) \mid X\right]$$

Marginalize $\rightsquigarrow$ $r_\Lambda(\delta) \geq r_\Lambda(\delta_\Lambda) < \infty$ (take $\delta = \delta_0$)

($\Rightarrow$) Assume $r_\Lambda(\delta_\Lambda) < \infty$ (o.w. δ_0 better)

Define $E_X(d) = \mathbb{E}\left[L(\theta, d) \mid X\right]$

$$A_\varepsilon = \{x : E_x(\delta_\Lambda(x)) > \varepsilon + \inf_d E_x(d)\}$$

For some $\varepsilon > 0$, $\mathbb{P}(X \in A_\varepsilon) > 0$

On A_ε choose $\delta^*(x)$ so $E_x(\delta^*(x)) \leq E_x(\delta_\Lambda(x)) - \varepsilon$

$\delta^* = \delta_\Lambda$ on A_ε^c $\Rightarrow$ $E_X(\delta_\Lambda(x)) - E_X(\delta^*(x)) \geq \varepsilon 1\{x \in A_\varepsilon\}$

Take expectations $\rightsquigarrow$ $r_\Lambda(\delta_\Lambda) - r_\Lambda(\delta^*) \geq \varepsilon \mathbb{P}(X \in A_\varepsilon)$ $\boxtimes$

Posterior Mean

If $L(\theta, d) = (g(\theta) - d)^2$ then the Bayes estimator is the posterior mean:

$$\mathbb{E}[(g(\theta)-d)^2 \mid X]$$
$$= \mathbb{E}\left[\left(g(\theta) - \mathbb{E}[g(\theta) \mid X] + \mathbb{E}[g(\theta) \mid X] - d\right)^2 \mid X\right]$$
$$= \mathrm{Var}(g(\theta) \mid X) + \left(\mathbb{E}[g(\theta) \mid X] - d\right)^2$$

(why is the cross-term 0?)

$$\Rightarrow \delta_\Lambda(x) = \mathbb{E}[g(\theta) \mid X = x]$$

Weighted sq. error:

$$L(\theta, d) = w(\theta)(g(\theta) - d)^2 \qquad \text{e.g. } \left(\frac{\theta - d}{\theta}\right)^2$$

sq. rel. error

$$\mathbb{E}[(d - g(\theta))^2 w(\theta) \mid X]$$
$$= d^2\, \mathbb{E}[w(\theta) \mid X] - 2d\, \mathbb{E}[w(\theta) g(\theta) \mid X]$$
$$+ \underbrace{\mathbb{E}[w(\theta) g(\theta)^2 \mid X]}_{\text{no dep. on } d}$$

min at $d = \dfrac{\mathbb{E}[w(\theta) g(\theta) \mid X]}{\mathbb{E}[w(\theta) \mid X]} \quad (= \delta_\Lambda(x))$

Example: Beta-Binomial

$$X \mid \theta \sim \text{Binom}(n, \theta) = \theta^x (1-\theta)^{n-x} \binom{n}{x}$$

$$\theta \sim \text{Beta}(\alpha, \beta) = \theta^{\alpha-1}(1-\theta)^{\beta-1} \underbrace{\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}}_{\text{normalizing const.}}$$

θ is r.v. here

Marginal dist. of X called <u>Beta-Binomial</u>

<u>Posterior:</u>

$$\lambda(\theta \mid x) = \lambda(\theta)\, p_\theta(x) / q(x)$$

we can drop factors that don't depend on θ

$$\propto_\theta \theta^{\alpha-1}(1-\theta)^{\beta-1}\, \theta^x (1-\theta)^{n-x}$$

$$= \theta^{x+\alpha-1}(1-\theta)^{n-x+\beta-1}$$

$$\Rightarrow \theta \mid X=x \sim \text{Beta}(x+\alpha,\, n-x+\beta)$$

$$\mathbb{E}[\theta \mid X] = \frac{X+\alpha}{n+\alpha+\beta}$$

convex combo of $\frac{X}{n}$, $\frac{\alpha}{\alpha+\beta}$

$$= \underbrace{\frac{X}{n}}_{\text{UMVU}} \cdot \frac{n}{n+\alpha+\beta} + \underbrace{\frac{\alpha}{\alpha+\beta}}_{\text{Prior Expectation}} \cdot \underbrace{\frac{\alpha+\beta}{n+\alpha+\beta}}_{\left(1-\frac{n}{n+\alpha+\beta}\right)}$$

<u>Interp.:</u> $k = \alpha+\beta$ "pseudo-trials," α successes

(Recall $\frac{X+3}{n+6}$ from Lec. 2)

Example: Normal mean

$$X|\theta \sim N(\theta, \sigma^2) \propto_\theta e^{-(x-\theta)^2/2\sigma^2}$$

$$\sim N(\mu, \tau^2) \propto_\theta e^{-(\theta-\mu)^2/2\tau^2}$$

$$\lambda(\theta|x) \propto_\theta \exp\left\{ -\frac{(x-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu)^2}{2\tau^2} \right\}$$

$$\propto_\theta \exp\left\{ \frac{x\theta}{\sigma^2} - \frac{\theta^2}{2\sigma^2} - \frac{\theta^2}{2\tau^2} + \frac{\theta\mu}{\tau^2} \right\}$$

$$= \exp\left\{ \theta \underbrace{\left(\frac{x}{\sigma^2} + \frac{\mu}{\tau^2}\right)}_{b} - \theta^2 \underbrace{\left(\frac{\sigma^{-2}+\tau^{-2}}{2}\right)}_{a^2} \right\}$$

Complete square:

$$a^2\theta^2 - b\theta = \left(\theta a - \frac{b}{2a}\right)^2 - c(a,b)$$

$$= \left(\theta - \frac{b}{2a^2}\right)^2 \cdot a^2 - c(a,b)/a$$

$$\propto_\theta \exp\left\{ -\left(\theta - \frac{x\sigma^{-2} + \mu\tau^{-2}}{\sigma^{-2}+\tau^{-2}}\right)^2 \Big/ 2\left(\sigma^{-2}+\tau^{-2}\right)^{-1} \right\}$$

$$\propto_\theta N\left(\underbrace{\frac{x\sigma^{-2} + \mu\tau^{-2}}{\sigma^{-2}+\tau^{-2}}}_{\text{precision-weighted average of } x,\, \mu}, \frac{1}{\sigma^{-2}+\tau^{-2}} \right)$$

$$\mathbb{E}[\theta | x] = x \cdot \frac{\sigma^{-2}}{\sigma^{-2}+\tau^{-2}} + \mu \cdot \frac{\tau^{-2}}{\sigma^{-2}+\tau^{-2}}$$

Gaussian iid sample

$$\Theta \sim N(\mu, \tau^2), \quad X_i \mid \theta \overset{iid}{\sim} N(\theta, \sigma^2), \quad i = 1, \dots, n$$

Suff. reduction $\rightsquigarrow$ $\bar{X} \mid \theta \sim N(\theta, \frac{\sigma^2}{n})$

$$\Rightarrow \mathbb{E}[\Theta \mid X] = \bar{X} \cdot \frac{n\sigma^{-2}}{n\sigma^{-2} + \tau^{-2}} + \mu \cdot \frac{\tau^{-2}}{n\sigma^{-2} + \tau^{-2}}$$

$$= \bar{X} \cdot \frac{n}{n + \sigma^2/\tau^2} + \mu \cdot \frac{\sigma^2/\tau^2}{n + \sigma^2/\tau^2}$$

<u>Interp</u>: $k = \sigma^2/\tau^2$ pseudo-observations, mean μ

If $n \gg k$, "data swamps prior"

If $n \ll k$, "prior swamps data"

[Note in both examples:

- Prior & Likelihood have similar fcn. form
- Posterior comes from same exp. fam. as prior]

If the posterior is from the same family as the prior, we say the prior is <u>conjugate</u> to the likelihood.

Most common in exp. families

Conjugate Priors

Suppose

$$X_i \mid \eta \overset{iid}{\sim} p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x) \qquad \eta \in \Xi \subseteq \mathbb{R}^s, \quad i = 1, \ldots, n$$

For carrier $\lambda_0(\eta)$, define $s+1$-dim family:

$$\lambda_{\mu,k}(\eta) = e^{k\mu'\eta - kA(\eta) - B(k\mu, k)} \lambda_0(\eta)$$

Suff. stat $\begin{pmatrix} \eta \\ -A(\eta) \end{pmatrix} \in \mathbb{R}^{s+1}$ Nat. param. $\begin{pmatrix} k\mu \\ k \end{pmatrix}$

$$\Rightarrow \lambda(\eta \mid x_1, \ldots, x_n) \propto_\eta \left(\prod_{i=1}^n e^{\eta' T(x_i) - A(\eta)} h(x_i) \right) \cdot e^{k\mu'\eta - kA(\eta) - B(k\mu, k)} \lambda_0(\eta)$$

$$\propto_\eta e^{(k\mu + \Sigma T(x_i))'\eta - (k+n)A(\eta)} \lambda_0(\eta)$$

$$= \lambda_{\mu_{post},\, k+n}(\eta)$$

where

$$\mu_{post} = \frac{k\mu + n\bar{T}}{k+n}, \quad \bar{T}(X) = \frac{1}{n}\sum_{i=1}^n T(x_i)$$

often Bayes est. for $\mathbb{E}_\eta T$

then

$$\mu_{post} = \bar{T} \cdot \frac{n}{k+n} + \mu \cdot \frac{k}{k+n}$$

UMVUE from data

"UMVUE" from "pseudo data"

Conjugate Prior Examples

Likelihood	Prior
$X_i \mid \theta \sim \text{Binom}(n, \theta)$ $= \theta^x (1-\theta)^{n-x} \binom{n}{x}$	$\theta \sim \text{Beta}(\alpha, \beta)$ $= \theta^{\alpha-1}(1-\theta)^{\beta-1} \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
$X_i \mid \theta \sim N(\theta, \sigma^2)$ (σ^2 known) $= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(\theta-x)^2/2\sigma^2}$	$\theta \sim N(\mu, \tau^2)$ $= \frac{1}{\sqrt{2\pi\tau^2}} e^{(\theta-\mu)^2/2\tau^2}$
$X_i \mid \theta \sim \text{Pois}(\theta)$ $x = 0, 1, \ldots$ $= \frac{\theta^x e^{-\theta}}{x!}$	$\theta \sim \text{Gamma}(v, s)$ $\theta > 0$ $= \frac{1}{\Gamma(v) s^2} \theta^{v-1} e^{-\theta/s}$

Gamma / Poisson:

$$\lambda(\theta \mid x) \propto_\theta \theta^{v-1+\Sigma x_i} e^{-(s^{-1}+n)\theta}$$

$$= \text{Gamma}(v + \Sigma x_i, (s^{-1}+n)^{-1})$$

$\Rightarrow k = s^{-1}, \quad \mu = vs$

$\lambda_0(\theta) = \theta^{-1}$ (not normalizable)

Flexibility of Bayes

Any Λ, $\mathcal{P}$, L, $g(\theta)$: δ_Λ defined straightforwardly

$$\delta_\Lambda(x) = \arg\min_d \int L(\theta, d)\, \lambda(\theta | x)\, d\theta$$

Problem reduced to (possibly hard) computation

Posterior is "one stop shop" for all answers

No need for:

- special family structure (exp. fam. / complete s.s.)
- special estimator (u-estimable)
- convex or nice L

$\Rightarrow$ Highly expressive modeling & estimation

Caveat: Limited by ability to do computations